100 Real-Life Developer Challenges

Master These to Match 5 Years of Professional Experience

> **Philosophy**: These aren't tutorial projects. These are actual problems developers face daily in production environments. Complete these and you'll have battle-tested skills.

🎯 How to Use This List

- ✅ Don't just build - **debug, optimize, and refactor**
- ✅ Write tests for your code
- ✅ Handle edge cases and errors properly
- ✅ Document your code
- ✅ Use version control professionally (meaningful commits)
- ✅ Deploy to production

SECTION 1: Frontend Mastery (1-25)

Level 1: UI Engineering Challenges

Challenge 1: Complex Form Wizard

The Problem: Build a multi-step form with validation, conditional fields, and data persistence across page refreshes.

- Form with 5+ steps
- Field validation (sync + async like username availability)
- Save progress to localStorage
- Back/forward navigation without losing data
- Handle form abandonment
- Show progress indicator

Skills: State management, form handling, localStorage, UX patterns

---

Challenge 2: Infinite Scroll with Virtual List

The Problem: Display 10,000+ items without crashing the browser.

- Only render visible items (virtualization)
- Smooth scrolling performance
- Handle variable height items
- Maintain scroll position on data update
- Add search/filter without lag

Skills: Performance optimization, DOM manipulation, intersection observer

---

Challenge 3: Drag & Drop Kanban Board

The Problem: Build Trello-like board from scratch (no libraries).

- Drag cards between columns
- Reorder cards within columns
- Touch device support
- Keyboard accessibility
- Optimistic UI updates
- Undo/redo functionality

Skills: Drag & Drop API, touch events, accessibility, state management

---

Challenge 4: Real-time Collaborative Text Editor

The Problem: Multiple users editing same document simultaneously.

- Cursor positions of other users
- Conflict resolution
- Offline support with sync
- Change history
- User presence indicators

Skills: WebSockets, CRDT/OT concepts, offline-first architecture

---

Challenge 5: Custom Design System

The Problem: Build a reusable component library from scratch.

- 15+ components (Button, Input, Modal, Dropdown, Toast, Table, etc.)
- Consistent theming (dark/light mode)
- Responsive by default
- Accessible (WCAG 2.1 AA)
- TypeScript support
- Storybook documentation

Skills: Component architecture, CSS architecture, accessibility, documentation

---

Challenge 6: Complex Data Table

The Problem: Enterprise-grade data table like Excel.

- Sorting (multi-column)
- Filtering (multiple conditions)
- Pagination + infinite scroll option
- Column resizing & reordering
- Row selection (single, multi, range)
- Inline editing
- Export to CSV/Excel
- Fixed headers & columns

Skills: Performance, complex state, UX patterns

Challenge 7: Image Editor

The Problem: Browser-based image editor.

- Crop, resize, rotate
- Filters (brightness, contrast, saturation)
- Text overlay
- Drawing tools
- Undo/redo stack
- Export in multiple formats
- Handle large images without freezing

Skills: Canvas API, Web Workers, file handling

Challenge 8: Animation System

The Problem: Build smooth, performant animations.

- Page transition animations
- Scroll-triggered animations
- Staggered list animations
- Gesture-based animations (swipe, pinch)
- Spring physics animations
- 60fps on mobile devices

Skills: CSS animations, JavaScript animations, requestAnimationFrame, FLIP technique

---

Challenge 9: Offline-First PWA

The Problem: App works perfectly without internet.

- Service worker caching strategies
- Background sync for pending actions
- IndexedDB for data storage
- Conflict resolution when back online
- Push notifications
- Install prompt handling

Skills: Service Workers, IndexedDB, PWA APIs

Challenge 10: Accessibility Audit & Fix

The Problem: Make an inaccessible app fully accessible.

- Screen reader testing
- Keyboard navigation
- Focus management
- ARIA labels & roles
- Color contrast fixes
- Reduced motion support
- Document your findings & fixes

Skills: Accessibility, testing with assistive technology

Level 2: State & Data Challenges

Challenge 11: State Management from Scratch

The Problem: Build your own Redux/Zustand.

- Global state container
- Actions & reducers pattern
- Middleware support
- DevTools integration
- React hooks integration
- Persistence layer

Skills: Design patterns, JavaScript fundamentals, debugging

Challenge 12: Real-time Dashboard

The Problem: Dashboard with live updating charts.

- 5+ different chart types
- WebSocket data streaming
- Configurable refresh rates
- Data aggregation (hourly, daily, weekly)
- Export functionality
- Handle connection drops gracefully

Skills: Data visualization, real-time data, charting libraries

---

Challenge 13: Search with Autocomplete

The Problem: Google-like search experience.

- Debounced search
- Keyboard navigation through results
- Recent searches
- Search suggestions
- Highlight matching text
- Handle slow network gracefully
- Cancel pending requests on new search

Skills: Debouncing, API optimization, UX patterns

Challenge 14: Complex Filtering System

The Problem: E-commerce style filtering.

- Multiple filter types (checkbox, range, date)
- URL state sync (shareable filtered URLs)
- Clear individual filters
- Show active filter count
- Handle filter combinations
- Performance with 1000+ products

Skills: URL state management, query strings, performance

Challenge 15: Undo/Redo System

The Problem: Implement command pattern for any app.

- Support arbitrary actions
- Memory-efficient history
- Branching history (tree structure)
- Serialize/deserialize history
- Keyboard shortcuts

Skills: Command pattern, data structures, memory management

Level 3: Integration Challenges

Challenge 16: Payment Integration

The Problem: Complete checkout flow.

- Stripe/Razorpay integration
- Card validation
- 3D Secure handling
- Webhook processing
- Invoice generation
- Subscription handling
- Handle failed payments

Skills: Payment APIs, security, error handling

Challenge 17: OAuth Implementation

The Problem: Social login that actually works.

- Google, GitHub, Facebook login
- Account linking
- Handle edge cases (email already exists)
- Session management
- Token refresh
- Proper logout (revoke tokens)

Skills: OAuth 2.0, security, session management

Challenge 18: File Upload System

The Problem: Handle files like a pro.

- Drag & drop upload
- Progress indicators
- Chunked upload for large files
- Resume interrupted uploads
- Image preview before upload
- File type validation (real validation, not just extension)
- Virus scanning integration
- S3/Cloudinary upload

Skills: File APIs, streaming, cloud storage

Challenge 19: Maps Integration

The Problem: Location-based features.

- Display markers on map
- Clustering for many markers
- Draw routes
- Geofencing
- Address autocomplete
- Reverse geocoding
- Handle location permissions
- Offline map caching

Skills: Maps APIs, geolocation, performance

Challenge 20: Video Player

The Problem: Custom video player.

- Custom controls
- Quality selection
- Playback speed
- Picture-in-picture
- Keyboard shortcuts
- Subtitles/captions
- Chapters/markers
- Analytics (watch time, engagement)

Skills: Video API, streaming, accessibility

Challenge 21: Email Template Builder

The Problem: Drag & drop email designer.

- Drag & drop blocks
- Rich text editing
- Variable placeholders
- Preview across email clients
- Export to HTML
- Template library
- Handle email client quirks

Skills: Drag & drop, HTML email limitations, WYSIWYG

Challenge 22: PDF Generation

The Problem: Generate professional PDFs.

- Invoice generation
- Report generation with charts
- Table of contents
- Headers/footers
- Page numbers
- Handle page breaks properly
- Custom fonts
- Digital signatures

Skills: PDF libraries, server-side rendering

Challenge 23: Calendar/Scheduling

The Problem: Calendly-like scheduling.

- Month/week/day views
- Event creation & editing
- Recurring events
- Timezone handling
- Availability slots
- Conflict detection
- Calendar sync (Google Calendar)

Skills: Date/time handling, timezones, external APIs

Challenge 24: Notification System

The Problem: Multi-channel notifications.

- In-app notifications
- Email notifications
- Push notifications
- SMS notifications
- Notification preferences
- Read/unread status
- Notification grouping
- Rate limiting

Skills: Push APIs, email services, message queues

Challenge 25: Chat System

The Problem: Real-time messaging.

- 1:1 and group chats
- Typing indicators
- Read receipts
- File sharing
- Emoji reactions
- Message search

- Online status
- Message encryption

Skills: WebSockets, real-time architecture, encryption

SECTION 2: Backend Mastery (26-50)

Level 1: API Design & Development

Challenge 26: RESTful API Design

The Problem: Design API for a complex domain.

- Proper resource naming
- HTTP methods usage
- Status codes
- Pagination (cursor vs offset)
- Filtering & sorting
- API versioning
- HATEOAS links
- OpenAPI documentation

Skills: REST principles, API design, documentation

Challenge 27: GraphQL API

The Problem: Build efficient GraphQL server.

- Schema design
- Resolvers
- N+1 query problem solution (DataLoader)
- Authentication & authorization
- Subscriptions for real-time
- Rate limiting
- Query complexity analysis
- Caching

Skills: GraphQL, query optimization, real-time

Challenge 28: Authentication System

The Problem: Secure auth from scratch.

- JWT implementation
- Refresh token rotation
- Password hashing (bcrypt)
- Email verification
- Password reset flow
- Rate limiting login attempts
- Session management
- Device tracking

Skills: Security, cryptography, session management

Challenge 29: Authorization System

The Problem: Role-based + attribute-based access.

- RBAC implementation
- Permission hierarchies
- Resource-level permissions
- API endpoint protection
- Frontend route protection
- Audit logging
- Super admin capabilities

Skills: Authorization patterns, security

Challenge 30: Rate Limiting & Throttling

The Problem: Protect your API.

- Token bucket implementation
- Sliding window algorithm
- Per-user limits
- Per-endpoint limits
- IP-based limiting
- Graceful degradation
- Rate limit headers
- Redis-based distributed limiting

Skills: Algorithms, Redis, API protection

Challenge 31: Caching Strategy

The Problem: Make your app fast.

- In-memory caching
- Redis caching
- Cache invalidation strategies
- Cache warming
- Cache stampede prevention
- ETags implementation
- CDN caching
- Query result caching

Skills: Caching patterns, Redis, performance

Challenge 32: Background Jobs

The Problem: Handle async tasks.

- Job queue implementation
- Job scheduling (cron)
- Job retries with backoff
- Dead letter queue
- Job prioritization
- Concurrent job processing
- Job monitoring & alerts

Skills: Message queues, async processing

Challenge 33: Webhook System

The Problem: Send & receive webhooks.

- Webhook endpoint creation
- Signature verification
- Retry logic with backoff
- Webhook logs
- Rate limiting
- Batch webhook processing

- Webhook testing tools

Skills: HTTP, security, reliability

Challenge 34: Search Implementation

The Problem: Fast, relevant search.

- Full-text search
- Fuzzy matching
- Faceted search
- Search suggestions
- Search analytics
- Elasticsearch/Algolia integration
- Search result ranking
- Highlight matching terms

Skills: Search engines, relevance tuning

Challenge 35: Logging & Monitoring

The Problem: Know what's happening in production.

- Structured logging
- Log aggregation
- Error tracking (Sentry)
- APM setup
- Custom metrics
- Alerting rules
- Dashboard creation
- Log retention policies

Skills: Observability, monitoring tools

Level 2: Database & Data Challenges

Challenge 36: Database Schema Design

The Problem: Design schema for complex app.

- Entity relationships
- Normalization decisions
- Indexing strategy
- Soft deletes
- Audit trails
- Multi-tenancy
- Schema migrations
- Data archival

Skills: Database design, SQL

Challenge 37: Query Optimization

The Problem: Fix slow queries.

- EXPLAIN analysis
- Index optimization
- Query rewriting
- N+1 detection and fix
- Connection pooling
- Read replicas usage
- Query caching
- Slow query logging

Skills: Database performance, profiling

Challenge 38: Data Migration

The Problem: Migrate data without downtime.

- Zero-downtime migration
- Data transformation
- Rollback strategy
- Data validation
- Progress tracking
- Handle large datasets
- Dual-write pattern

Skills: Database operations, reliability

Challenge 39: Multi-tenancy

The Problem: One app, many customers.

- Tenant isolation
- Database per tenant vs shared
- Tenant-specific customization
- Cross-tenant queries (admin)
- Tenant onboarding/offboarding
- Billing per tenant

Skills: Architecture, security, scalability

Challenge 40: Event Sourcing

The Problem: Store events, not state.

- Event store implementation
- Event replay
- Snapshots for performance
- Event versioning
- CQRS pattern
- Eventual consistency handling

Skills: Architecture patterns, data modeling

Challenge 41: Data Sync

The Problem: Keep data in sync across systems.

- Real-time sync
- Conflict resolution
- Offline changes sync
- Sync status tracking
- Partial sync
- Delta sync optimization

Skills: Distributed systems, conflict resolution

Challenge 42: Backup & Recovery

The Problem: Never lose data.

- Automated backups
- Point-in-time recovery
- Backup verification
- Cross-region backups
- Recovery testing
- RTO/RPO planning
- Disaster recovery drill

Skills: Database operations, reliability

Level 3: Infrastructure & DevOps

Challenge 43: CI/CD Pipeline

The Problem: Automate everything.

- Automated testing
- Build optimization
- Deployment automation
- Environment management
- Rollback automation
- Feature flags integration
- Security scanning
- Deployment notifications

Skills: DevOps, automation, testing

Challenge 44: Docker & Containerization

The Problem: Consistent environments.

- Multi-stage builds
- Docker Compose setup
- Volume management
- Network configuration
- Image optimization
- Container health checks
- Secrets management
- Local dev environment

Skills: Docker, containers

Challenge 45: Kubernetes Basics

The Problem: Container orchestration.

- Deployments & services
- ConfigMaps & secrets
- Ingress configuration
- Horizontal scaling
- Health checks
- Rolling updates
- Resource limits
- Local K8s development

Skills: Kubernetes, orchestration

Challenge 46: Cloud Infrastructure

The Problem: Deploy to cloud properly.

- Infrastructure as code (Terraform/Pulumi)
- VPC & networking
- Load balancer setup
- Auto-scaling
- CDN configuration
- SSL/TLS setup
- Cloud storage
- Cost optimization

Skills: Cloud platforms, IaC

Challenge 47: Service Health & Reliability

The Problem: Keep services running.

- Health check endpoints
- Graceful shutdown
- Circuit breakers
- Retry patterns
- Timeout handling
- Bulkhead pattern

- Chaos engineering basics

Skills: Reliability patterns, resilience

Challenge 48: Security Implementation

The Problem: Secure your application.

- HTTPS everywhere
- CORS configuration
- CSP headers
- SQL injection prevention
- XSS prevention
- CSRF protection
- Security headers
- Dependency vulnerability scanning
- Secrets management
- Penetration testing basics

Skills: Security, OWASP

Challenge 49: Performance Testing

The Problem: Know your limits.

- Load testing setup
- Stress testing
- Identify bottlenecks
- Performance baselines
- Automated performance tests
- Performance budgets
- Profiling tools

Skills: Performance, testing

Challenge 50: Microservices Communication

The Problem: Services talking to services.

- REST vs gRPC
- Service discovery

- API gateway
- Message queues
- Event-driven architecture
- Distributed tracing
- Service mesh basics

Skills: Distributed systems, architecture

SECTION 3: Full Stack Integration (51-75)

Challenge 51: E-commerce Platform

The Problem: Build complete shopping experience.

- Product catalog with variants
- Shopping cart (persistent)
- Checkout flow
- Inventory management
- Order management
- Shipping calculation
- Tax calculation
- Admin dashboard

Skills: Full stack, payments, complex state

Challenge 52: SaaS Starter Kit

The Problem: Foundation for SaaS products.

- Multi-tenant architecture
- Subscription billing
- User management
- Team/organization features
- Billing portal
- Usage tracking
- Admin super panel
- Onboarding flow

Skills: SaaS patterns, billing, multi-tenancy

Challenge 53: CMS from Scratch

The Problem: Content management system.

- Content types definition
- WYSIWYG editor
- Media library
- User roles
- Content versioning
- Preview functionality
- SEO features
- API for headless usage

Skills: CMS patterns, rich text, media handling

Challenge 54: Analytics Dashboard

The Problem: Track user behavior.

- Event tracking
- Session recording concept
- Funnel analysis
- Cohort analysis
- Custom dashboards
- Data export
- Real-time analytics
- Privacy compliance

Skills: Data processing, visualization, privacy

Challenge 55: Booking System

The Problem: Appointment/resource booking.

- Availability calendar
- Booking slots
- Conflict prevention
- Cancellation handling
- Reminders
- Waitlist
- Buffer times

- Resource management

Skills: Date/time, concurrency, notifications

Challenge 56: Social Platform Features

The Problem: Social networking features.

- User profiles
- Follow/unfollow
- News feed algorithm
- Like/comment system
- Content moderation
- Hashtags & mentions
- Privacy settings
- Block/report users

Skills: Social patterns, feeds, moderation

Challenge 57: Marketplace Platform

The Problem: Two-sided marketplace.

- Buyer/seller flows
- Listing management
- Search & discovery
- Messaging between parties
- Transaction management
- Escrow concept
- Reviews & ratings
- Dispute resolution

Skills: Marketplace patterns, trust & safety

Challenge 58: Learning Management System

The Problem: Online course platform.

- Course creation
- Video lessons
- Quizzes & assignments

- Progress tracking
- Certificates
- Discussion forums
- Enrollment management
- Instructor dashboard

Skills: Video streaming, progress tracking, gamification

Challenge 59: Project Management Tool

The Problem: Jira/Asana clone.

- Projects & tasks
- Kanban & list views
- Sprint planning
- Time tracking
- Team assignments
- Notifications
- Activity feed
- Integrations (GitHub, Slack)

Skills: Complex state, real-time, integrations

Challenge 60: Invoice & Billing System

The Problem: Business invoicing.

- Invoice creation
- Recurring invoices
- Payment tracking
- Late payment reminders
- Tax handling
- Multi-currency
- Reports
- Client portal

Skills: Financial calculations, PDF generation, automation

Challenge 61: HR Management System

The Problem: Employee management.

- Employee directory
- Leave management
- Attendance tracking
- Payroll basics
- Performance reviews
- Document management
- Onboarding checklist
- Org chart

Skills: Business logic, workflows, reporting

Challenge 62: Inventory Management

The Problem: Stock management.

- Product tracking
- Stock levels
- Low stock alerts
- Purchase orders
- Suppliers management
- Barcode/QR scanning
- Stock movement history
- Multi-location support

Skills: Business logic, real-time updates, hardware integration

---

Challenge 63: Customer Support System

The Problem: Help desk / ticketing.

- Ticket creation
- Assignment rules
- SLA tracking
- Canned responses
- Knowledge base
- Live chat integration
- Customer satisfaction surveys
- Agent performance metrics

Skills: Workflows, real-time, automation

Challenge 64: Event Management

The Problem: Event ticketing platform.

- Event creation
- Ticket types & pricing
- Seat selection
- Registration forms
- QR code tickets
- Check-in system
- Waitlist management
- Event analytics

Skills: Booking, QR codes, real-time capacity

Challenge 65: Restaurant System

The Problem: Restaurant management.

- Menu management
- Table reservations
- Order taking (dine-in)
- Online ordering
- Kitchen display system
- Bill splitting
- Loyalty program
- Inventory integration

Skills: Real-time, POS patterns, multiple interfaces

Challenge 66: Real Estate Platform

The Problem: Property listing.

- Property listings
- Advanced search filters
- Map-based search
- Virtual tours integration
- Agent profiles
- Inquiry management
- Saved searches

- Price alerts

Skills: Search, maps, media handling

Challenge 67: Healthcare Appointment

The Problem: Doctor appointment system.

- Doctor profiles
- Specialization filtering
- Appointment booking
- Video consultation
- Prescription management
- Medical history
- Insurance integration
- HIPAA considerations

Skills: Booking, video calls, privacy/compliance

Challenge 68: Fitness/Gym Platform

The Problem: Fitness tracking & classes.

- Workout logging
- Progress charts
- Class scheduling
- Trainer assignments
- Membership management
- Goal setting
- Social challenges
- Wearable integration concepts

Skills: Data visualization, gamification, IoT concepts

Challenge 69: Food Delivery Platform

The Problem: Delivery service.

- Restaurant listings
- Menu management
- Cart & checkout

- Real-time order tracking
- Driver assignment
- Delivery zones
- Rating system
- Promo codes

Skills: Real-time, geolocation, multi-user types

Challenge 70: Job Board

The Problem: Job listing platform.

- Job postings
- Applicant tracking
- Resume parsing concept
- Application workflow
- Employer dashboard
- Job alerts
- Company profiles
- Application analytics

Skills: Search, file processing, workflows

Challenge 71: Survey/Form Builder

The Problem: Typeform-like builder.

- Drag & drop form builder
- Multiple question types
- Conditional logic
- Response collection
- Analytics & reports
- Export responses
- Embed functionality
- Response validation

Skills: Form building, dynamic schemas, analytics

Challenge 72: URL Shortener (Advanced)

The Problem: Link management platform.

- Short URL creation
- Custom aliases
- QR code generation
- Click analytics
- Geo tracking
- Device tracking
- A/B testing links
- Expiring links
- Password protection

Skills: Analytics, URL handling, security

Challenge 73: Newsletter Platform

The Problem: Email marketing.

- Subscriber management
- Email composer
- Template library
- Campaign scheduling
- A/B testing
- Analytics (opens, clicks)
- Unsubscribe handling
- Bounce management

Skills: Email delivery, analytics, compliance

Challenge 74: Wiki/Documentation

The Problem: Knowledge base.

- Nested pages
- Markdown/rich editor
- Version history
- Search functionality
- Access permissions
- Comments/discussions
- Export options
- API documentation generation

Skills: Content management, search, versioning

Challenge 75: Audit Log System

The Problem: Track all changes.

- Capture all mutations
- User identification
- Before/after values
- Searchable logs
- Retention policies
- Compliance reports
- Tamper-proof storage
- Query interface

Skills: Logging, compliance, data integrity

SECTION 4: Advanced Patterns & Architecture (76-90)

Challenge 76: API Gateway

The Problem: Build your own API gateway.

- Request routing
- Rate limiting
- Authentication
- Request/response transformation
- Caching
- Circuit breaker
- Logging & metrics
- Multiple backend support

Skills: Architecture, networking, patterns

Challenge 77: Feature Flag System

The Problem: Control feature rollout.

- Flag management UI

- Percentage rollouts
- User targeting
- A/B testing integration
- Kill switches
- Flag dependencies
- Audit trail
- SDK for apps

Skills: Feature management, experimentation

Challenge 78: Plugin System

The Problem: Extensible architecture.

- Plugin API design
- Plugin loading
- Sandboxing
- Marketplace concept
- Plugin versioning
- Configuration UI
- Event hooks
- Security considerations

Skills: Architecture, extensibility, security

Challenge 79: Multi-Region Deployment

The Problem: Global application.

- Data replication
- Request routing
- Latency optimization
- Failover handling
- Data consistency
- Deployment strategy
- Region-specific features

Skills: Global architecture, reliability

Challenge 80: Real-time Collaboration

The Problem: Google Docs-like sync.

- Operational transformation or CRDT
- Cursor synchronization
- Presence awareness
- Conflict resolution
- History/versions
- Offline support
- Selective sync

Skills: Distributed systems, algorithms

Challenge 81: Workflow Engine

The Problem: Business process automation.

- Visual workflow builder
- Conditional branching
- Parallel execution
- Human tasks
- Timer events
- Error handling
- Workflow versioning
- Monitoring

Skills: State machines, process automation

Challenge 82: Recommendation Engine

The Problem: Personalized suggestions.

- Collaborative filtering basics
- Content-based filtering
- Hybrid approach
- Real-time updates
- A/B testing recommendations
- Explanation of recommendations
- Cold start handling

Skills: Algorithms, ML basics, personalization

Challenge 83: Data Pipeline

The Problem: ETL/data processing.

- Data extraction
- Transformation logic
- Loading to destination
- Scheduling
- Error handling
- Data validation
- Monitoring
- Incremental processing

Skills: Data engineering basics

Challenge 84: Distributed Lock

The Problem: Coordinate distributed systems.

- Lock acquisition
- Lock release
- Timeout handling
- Deadlock prevention
- Distributed lock with Redis
- Leader election
- Fencing tokens

Skills: Distributed systems, concurrency

Challenge 85: Message Queue Implementation

The Problem: Build a basic message queue.

- Publish/subscribe
- Queue management
- Message persistence
- Acknowledgments
- Dead letter handling
- Message ordering
- Consumer groups

Skills: Messaging patterns, persistence

Challenge 86: Blue-Green Deployment

The Problem: Zero-downtime deployments.

- Environment duplication
- Traffic switching
- Database migration handling
- Rollback capability
- Health verification
- Smoke testing
- Monitoring during switch

Skills: DevOps, deployment strategies

Challenge 87: Load Balancer Understanding

The Problem: Build a simple load balancer.

- Round-robin distribution
- Health checks
- Session stickiness
- Weight-based routing
- Connection draining
- SSL termination concept

Skills: Networking, distribution

Challenge 88: CDN Integration

The Problem: Global content delivery.

- Static asset serving
- Cache invalidation
- Edge functions concept
- Custom domains
- SSL management
- Performance optimization
- Cost optimization

Skills: CDN, caching, performance

Challenge 89: Idempotency Implementation

The Problem: Safe retry of operations.

- Idempotency keys
- Request deduplication
- Proper response caching
- Timeout handling
- Database constraints
- Distributed idempotency

Skills: Reliability, distributed systems

Challenge 90: Graceful Degradation

The Problem: Work when things fail.

- Fallback strategies
- Circuit breaker implementation
- Timeout handling
- Retry with backoff
- Partial responses
- User communication
- Recovery automation

Skills: Resilience, reliability

SECTION 5: Testing & Quality (91-100)

Challenge 91: Test Suite from Scratch

The Problem: Comprehensive testing.

- Unit tests
- Integration tests
- E2E tests
- Test organization
- Mocking strategies
- Test data management
- Coverage reports

- CI integration

Skills: Testing, quality assurance

Challenge 92: Performance Optimization

The Problem: Make it fast.

- Frontend profiling
- Backend profiling
- Database optimization
- Caching implementation
- Bundle optimization
- Image optimization
- API response optimization
- Lighthouse score 90+

Skills: Performance, profiling

Challenge 93: Error Handling Strategy

The Problem: Graceful error handling.

- Error classification
- User-friendly messages
- Error logging
- Error tracking
- Recovery strategies
- Error boundaries
- API error responses
- Retry logic

Skills: Error handling, UX

Challenge 94: Code Review Practice

The Problem: Review code like a senior.

- Find 20 real GitHub PRs
- Identify issues
- Suggest improvements

- Consider edge cases
- Security review
- Performance review
- Write constructive feedback

Skills: Code review, communication

Challenge 95: Refactoring Challenge

The Problem: Improve existing code.

- Take a messy open-source project
- Identify code smells
- Plan refactoring
- Implement improvements
- Ensure no regressions
- Document decisions
- Measure improvement

Skills: Refactoring, code quality

Challenge 96: Debug Production Issue

The Problem: Fix under pressure.

- Set up a buggy app intentionally
- Reproduce issues
- Use debugging tools
- Read logs effectively
- Fix without breaking other things
- Write postmortem
- Implement preventive measures

Skills: Debugging, incident response

Challenge 97: Documentation Excellence

The Problem: Document like a pro.

- README that sells
- API documentation

- Architecture diagrams
- Setup guides
- Troubleshooting guides
- Contributing guidelines
- Code comments strategy
- Decision records (ADRs)

Skills: Documentation, communication

Challenge 98: Accessibility Compliance

The Problem: WCAG 2.1 AA compliance.

- Audit existing app
- Fix all issues
- Automated testing
- Screen reader testing
- Keyboard navigation
- Color contrast
- Focus management
- Documentation

Skills: Accessibility

Challenge 99: Security Audit

The Problem: Find and fix vulnerabilities.

- OWASP Top 10 check
- Dependency audit
- Authentication review
- Authorization review
- Data validation
- Secrets management
- Security headers
- Penetration testing basics

Skills: Security

Challenge 100: Production Incident Simulation

The Problem: Handle outages professionally.

- Simulate database failure
- Simulate third-party API outage
- Simulate traffic spike
- Write runbooks
- Practice incident response
- Communication templates
- Postmortem practice
- On-call simulation

Skills: Operations, incident management

📊 Progress Tracker

Completion Status

Section	Challenges	Completed
Frontend Mastery	1-25	☐/25
Backend Mastery	26-50	☐/25
Full Stack Integration	51-75	☐/25
Advanced Patterns	76-90	☐/15
Testing & Quality	91-100	☐/10

Difficulty Progression

```
Challenges 1-25:  ▓▓▓
```

You will have:

- ✅ Built production-ready features
- ✅ Handled real-world edge cases
- ✅ Debugged complex issues
- ✅ Optimized for performance
- ✅ Implemented security best practices
- ✅ Written comprehensive tests
- ✅ Deployed to production environments
- ✅ Handled incidents and outages
- ✅ Worked with distributed systems
- ✅ Built for scale

This is equivalent to 5+ years of professional development experience compressed into focused practice.

"The difference between a junior and senior developer isn't years—it's the variety and complexity of problems they've solved."